Appendix

AI Applications on Model Build (Python Project)

It is to supplement more details to the Python Project on the Model Build

Credit Risk Data Attributes (source from Kaggle)

Target Variable

- **Loan_Status:** Binary indicator (1 default vs. 0 non-default).

Features

- **Person_Age:** Age of the borrower.
- **Person_Income:** Annual income.
- **Person_Home_Ownership:** Categories include RENT, OWN, MORTGAGE, OTHER.
- **Person_Emp_Length:** Years of employment.
- **Loan_Intent:** Purpose of the loan (e.g., EDUCATION, MEDICAL, HOMEIMPROVEMENT, etc.).
- **Loan_Amnt:** Requested loan amount.
- **Loan_Int_Rate:** Interest rate on the loan.
- **Loan_Percent_Income:** Ratio depicting the requested loan amount relative to income.
- **CB_Person_Default_on_File:** Historical marker (Yes/No) for previous defaults.
- **CB_Person_Cred_Hist_Length:** Length in years of credit history.

Derived Features (in Feature Engineering)

- **Credit_Age_Ratio:** This ratio measures the proportion of a person's entire life during which they have had a credit history.
- **Employment Stability:** This score measures the ratio of the time a person has been employed to their available working years.

Have a glance on the data

	person_age	person_income	person_home_ownership	person_emp_length	loan_intent	loan_amnt
0	22	59000	RENT	123.0	PERSONAL	35000
1	21	9600	OWN	5.0	EDUCATION	1000
2	25	9600	MORTGAGE	1.0	MEDICAL	5500
3	23	65500	RENT	4.0	MEDICAL	35000
4	24	54400	RENT	8.0	MEDICAL	35000

loan_int_rate	loan_status	loan_percent_income	cb_person_default_on_file	cb_person_cred_hist_length
16.02	1	0.59	Y	3
11.14	0	0.10	N	2
12.87	1	0.57	N	3
15.23	1	0.53	N	2
14.27	1	0.55	Y	4

Note: Loan_Grade is removed because from experience of HKMA's model review, it is not preferred to have feature which is a score or grade (i.e. not prefer model on model).

Model Build Processes

Data Cleansing

- Remove Duplicates
- Remove Missing Values
- Handling Outliers

Feature Engineering

- One-Hot Encoding for Categorical Features
- Standardization for Numeric Features
- Creating New Features

Model Training

- Decision Tree / Logistic Regression / Random Forest / XGBoost

Model Testing

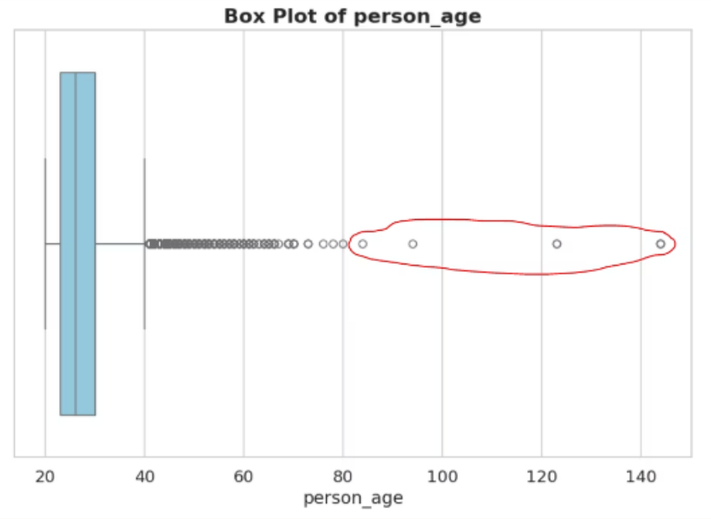
- AUC

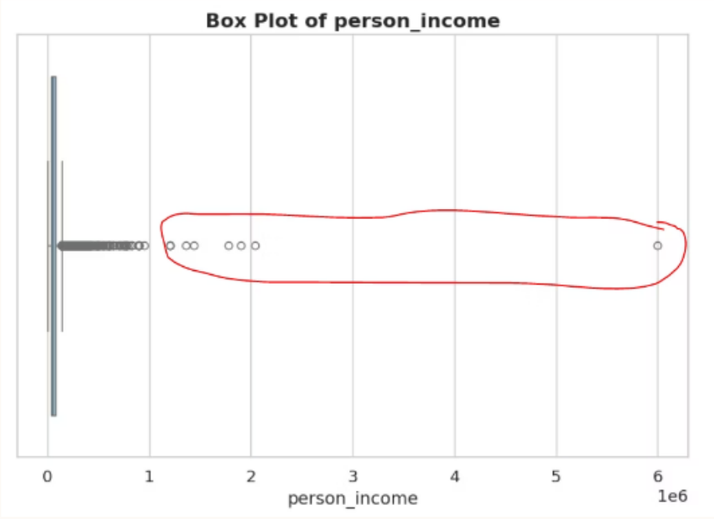
Model Explainability

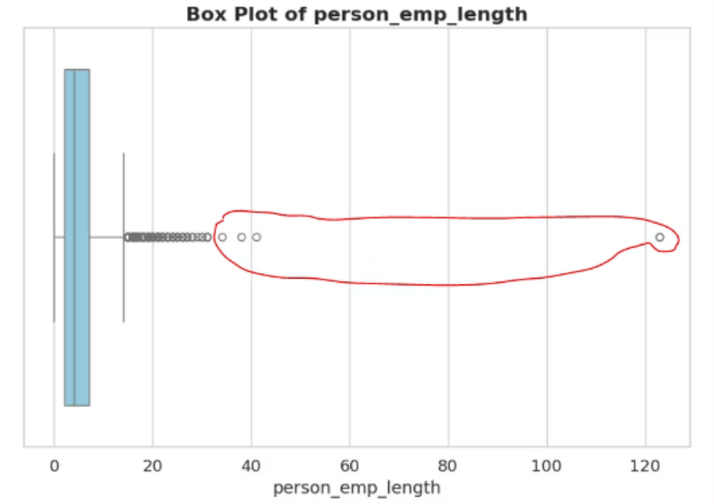
- SHAP

Exploratory Data Analysis

Numeric Features Overview







Person Age

Observation:

- Majority of applicants are young adults (median ~26).
- Presence of extreme outlier (144) suggests data entry errors.

Person Income

Observation:

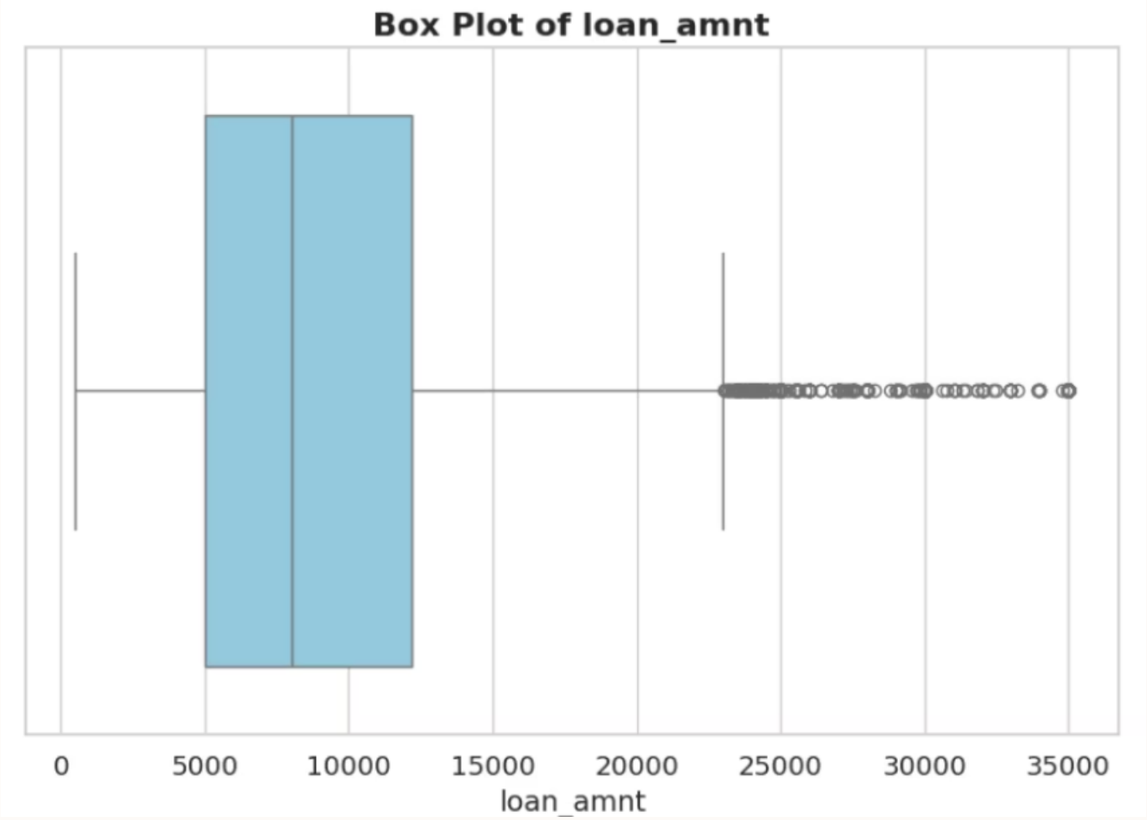
- Majority of incomes falling in a lower-to-mid range with right-skewed distribution
- A number of outlier points

Person Employment Length

Observation:

- Average of ~4.8 years; some unrealistic entries (max = 123 years) exist.
- Presence of extreme outlier (120) suggests data entry errors.

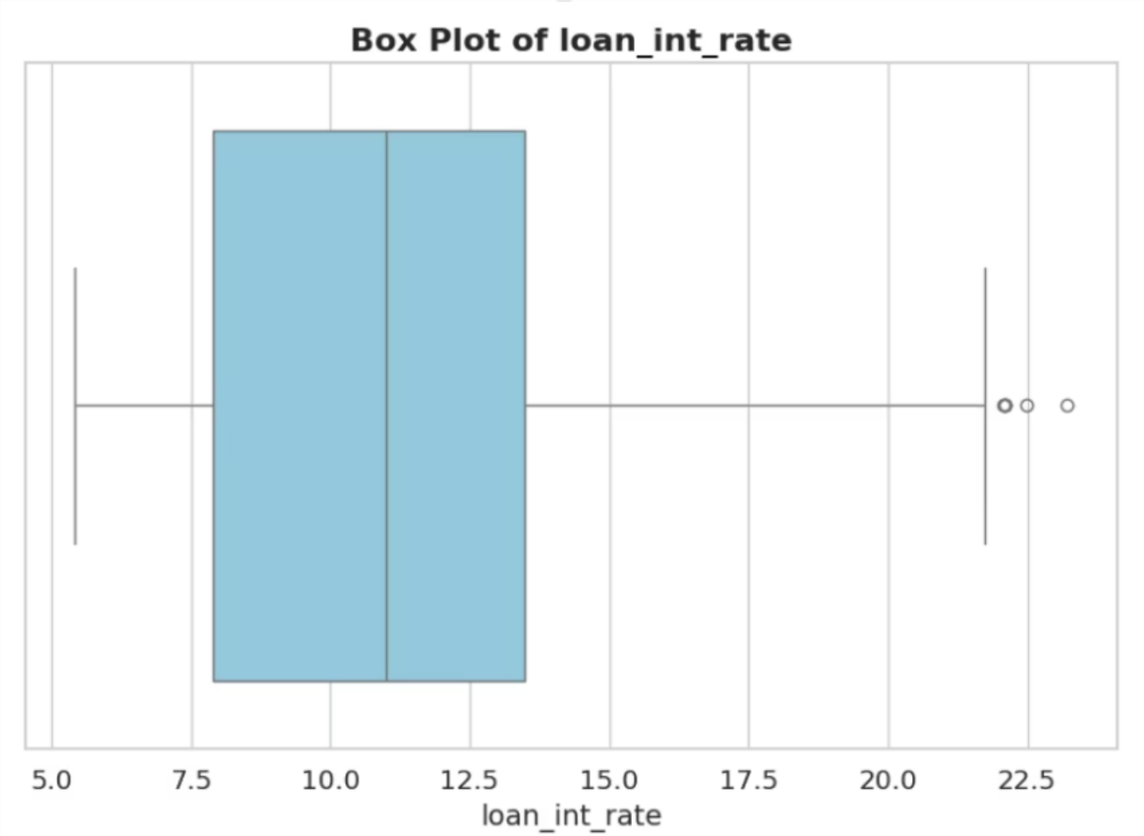
Numeric Features Overview (Continued)



Loan Amount

Observation:

- The distribution is moderately spread out, with most values centered around a median loan amount (approximately 8,000)
- Outliers may exist but are not as pronounced compared to income

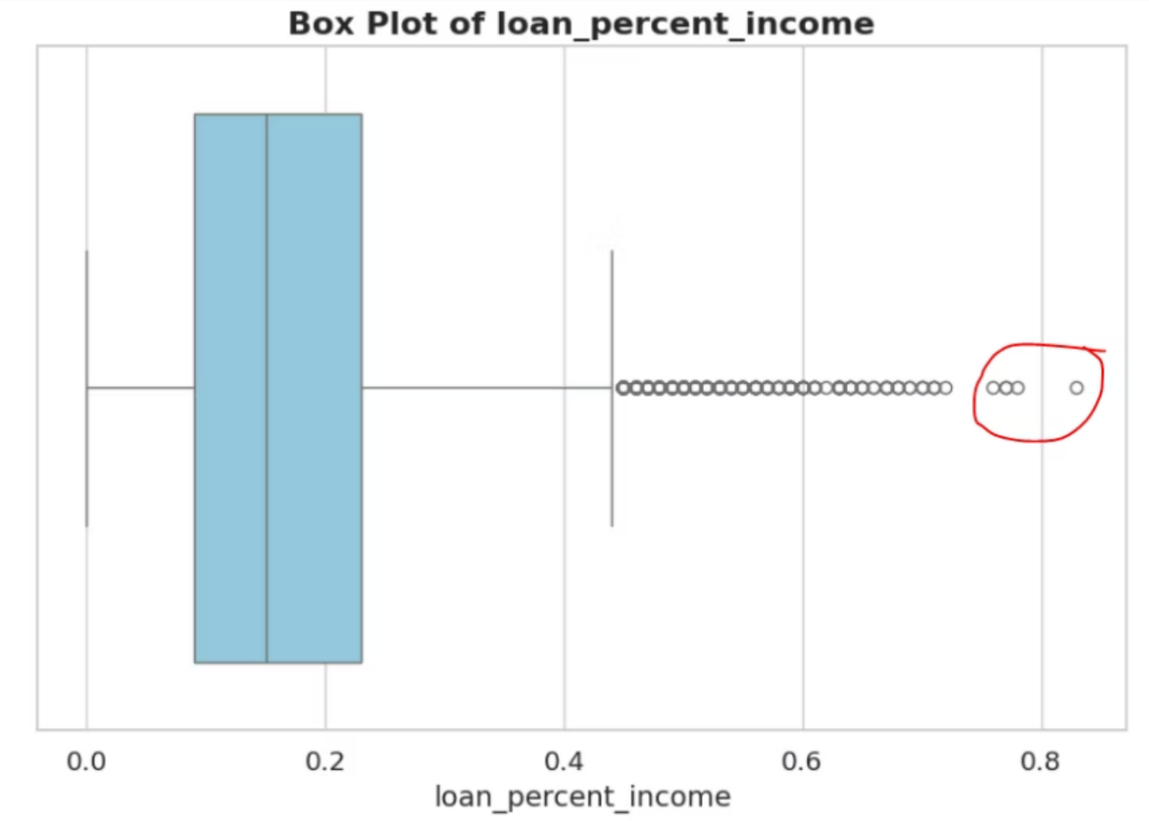


Loan Interest Rate

Observation:

- Average interest rate near 11%, with a relatively tight spread.
- Some higher values at the upper whisker suggest that certain loans have markedly higher interest rates.

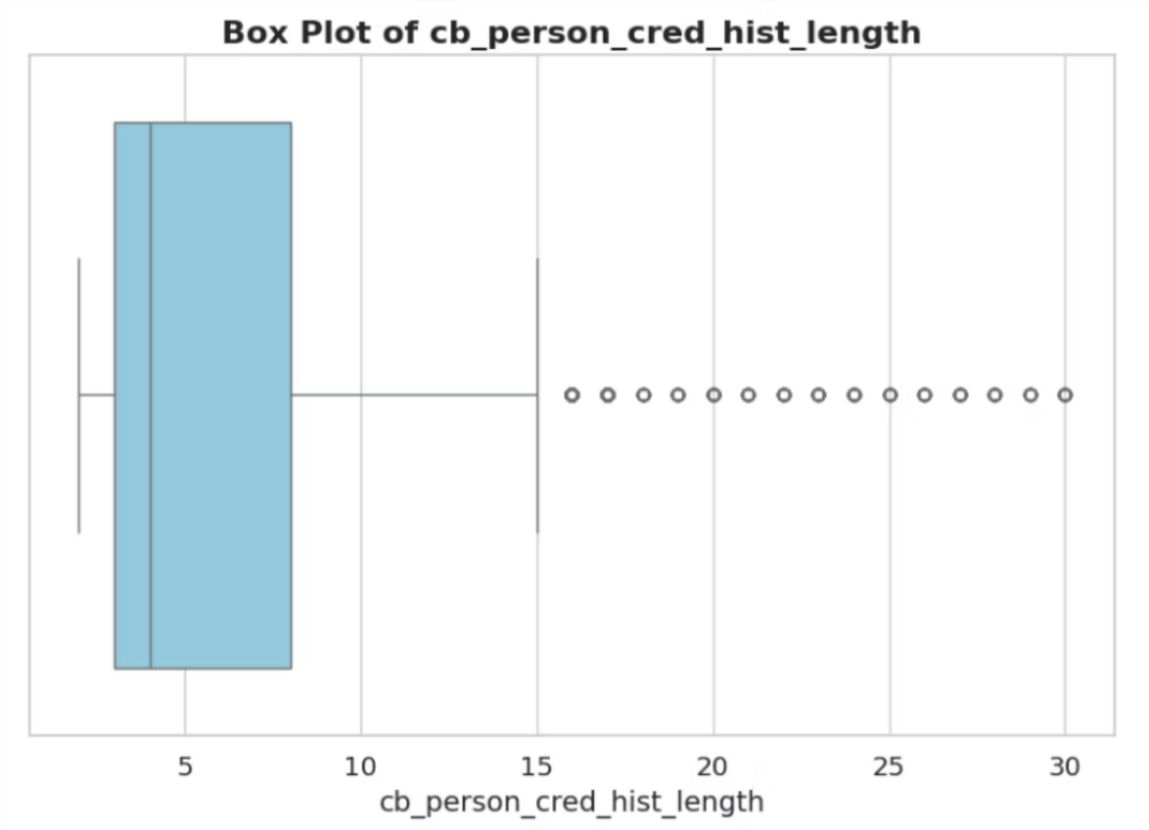
Numeric Features Overview (Continued)



Loan Percent Income

Observation:

- Represents the ratio of loan amount to income (median ~0.15)
- The high-end outliers suggest that for some, represents financial burden

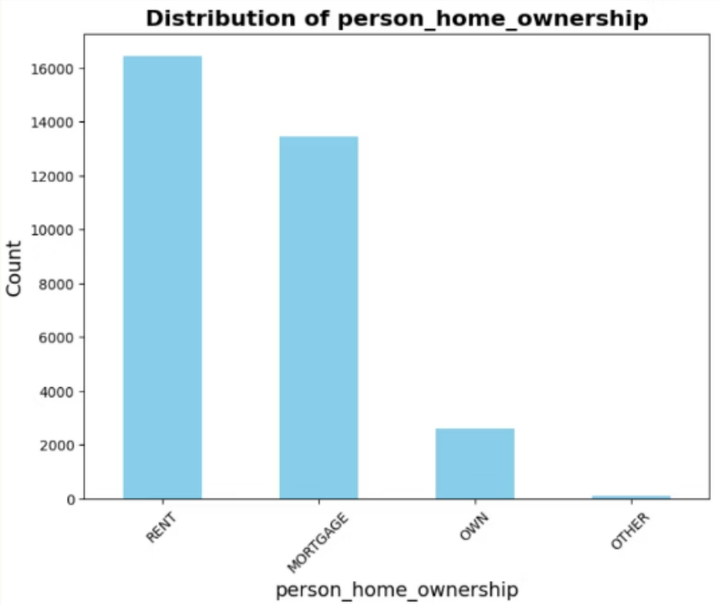


Credit History Length

Observation:

- Most borrowers have credit histories ranging from about 2 to 8 years, with a median around 4 years
- A long tail toward the higher end (up to 30 years) may identify a small subset with verv extended credit histories.

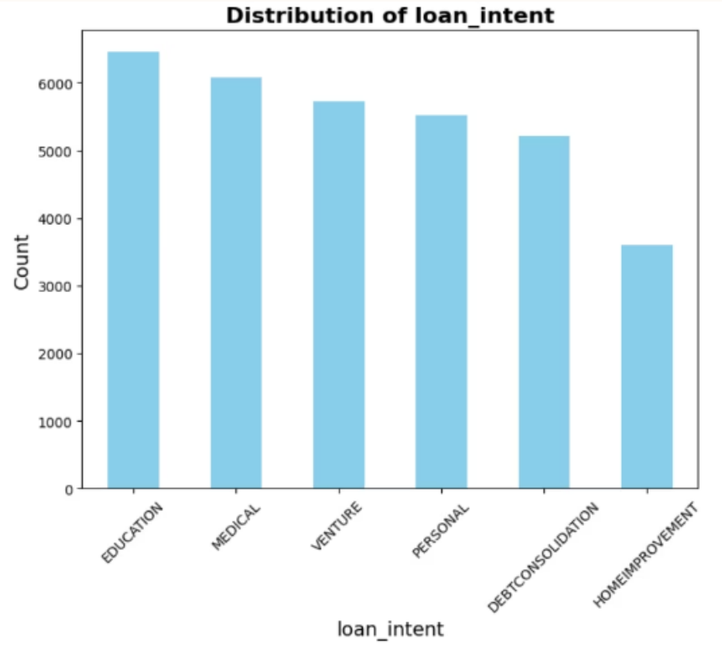
Categorical Features Overview



Person Home Ownership

Observation:

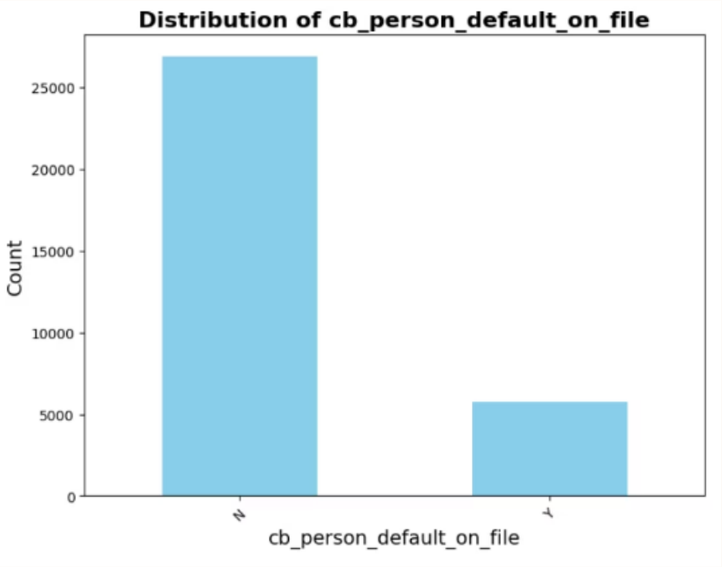
- Predominantly renters (≈ 16,446) and those with a mortgage (≈ 13,444).
- Serves as a proxy for financial stability.



Loan Intent

Observation:

- Applicants seek loans for diverse purposes (Education, Medical, Personal, etc.).

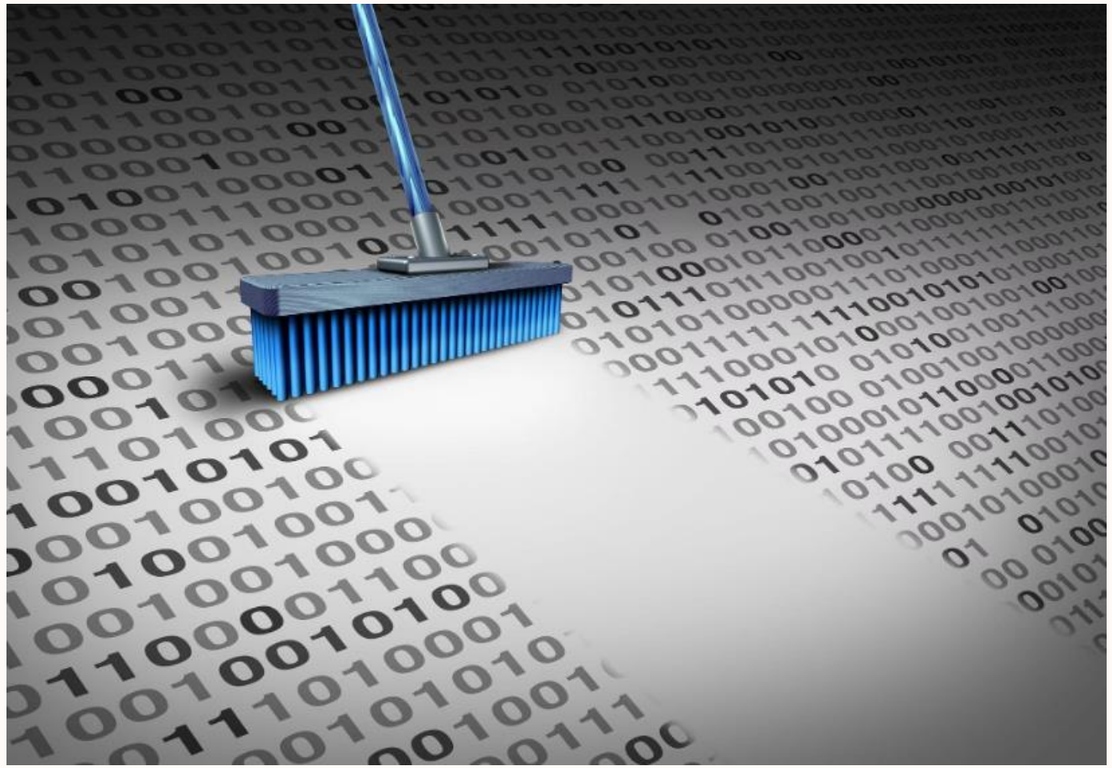


Credit Default History

Observation:

- Majority have no previous default (≈ 26,836 'N'), yet 17% show past defaults (≈ 5,745 'Y').

Data Cleansing



Remove Duplicates

There are 165 duplicates found which needed to be excluded from the dataset

Number of duplicate rows: 165

	person_age	person_income	person_home_ownership	person_emp_length	loan_intent	loan_amnt
15944	21	8088	RENT	NaN	MEDICAL	1200
16835	21	8088	RENT	NaN	MEDICAL	1200
2431	21	15600	RENT	0.0	MEDICAL	2800
17758	21	15600	RENT	0.0	MEDICAL	2800
2498	21	18000	RENT	0.0	DEBTCONSOLIDATION	3000

```
[12] # Removing duplicate rows
     fullDataRmDup = fullData.drop_duplicates()
```

```
[13] print("Before remove duplicates:")
     print(fullData.shape)
     print("After remove duplicates:")
     print(fullDataRmDup.shape)
```

```
Before remove duplicates:
(32581, 11)
After remove duplicates:
(32416, 11)
```

Remove Missing

- loan_int_rate and person_emp_length has missing values

- To preserve the data information, instead of remove the missing data, missing values were imputed with mean values

	person_age	person_income	person_home_ownership	person_emp_length	loan_int_rate
105	22	12600	MORTGAGE	NaN	NaN
222	24	185000	MORTGAGE	NaN	NaN
379	24	16800	MORTGAGE	NaN	NaN

```
fullData_no_missing['person_emp_length']
    = fullData_no_missing['person_emp_length'].fillna(fullData_no_missing['person_emp_length'].mean())
fullData_no_missing['loan_int_rate']
    = fullData_no_missing['loan_int_rate'].fillna(fullData_no_missing['loan_int_rate'].mean())
```

```
Check no. of NaN records before handling missing:
person_age                      0
person_income                   0
person_home_ownership           0
person_emp_length             887
loan_intent                     0
loan_amnt                       0
loan_int_rate                3095
loan_status                     0
loan_percent_income             0
cb_person_default_on_file       0
cb_person_cred_hist_length      0
dtype: int64
```

```
Check no. of NaN records after handling missing:
person_age                      0
person_income                   0
person_home_ownership           0
person_emp_length               0
loan_intent                     0
loan_amnt                       0
loan_int_rate                   0
loan_status                     0
loan_percent_income             0
cb_person_default_on_file       0
cb_person_cred_hist_length      0
dtype: int64
```

Handling Outliers

Mean ± 3σ (Preferred)

- Remove outliers with values out of *Mean ± 3σ*
- Captures 99.7% of the data, removing only extreme anomalies

IQR x 1.5

- Remove outliers with values out of *Q1 - IQR x 1.5 and Q3 + IQR x 1.5*
- The “fences” can be too tight→ risking the removal of valid data (removed around 40% of training samples)

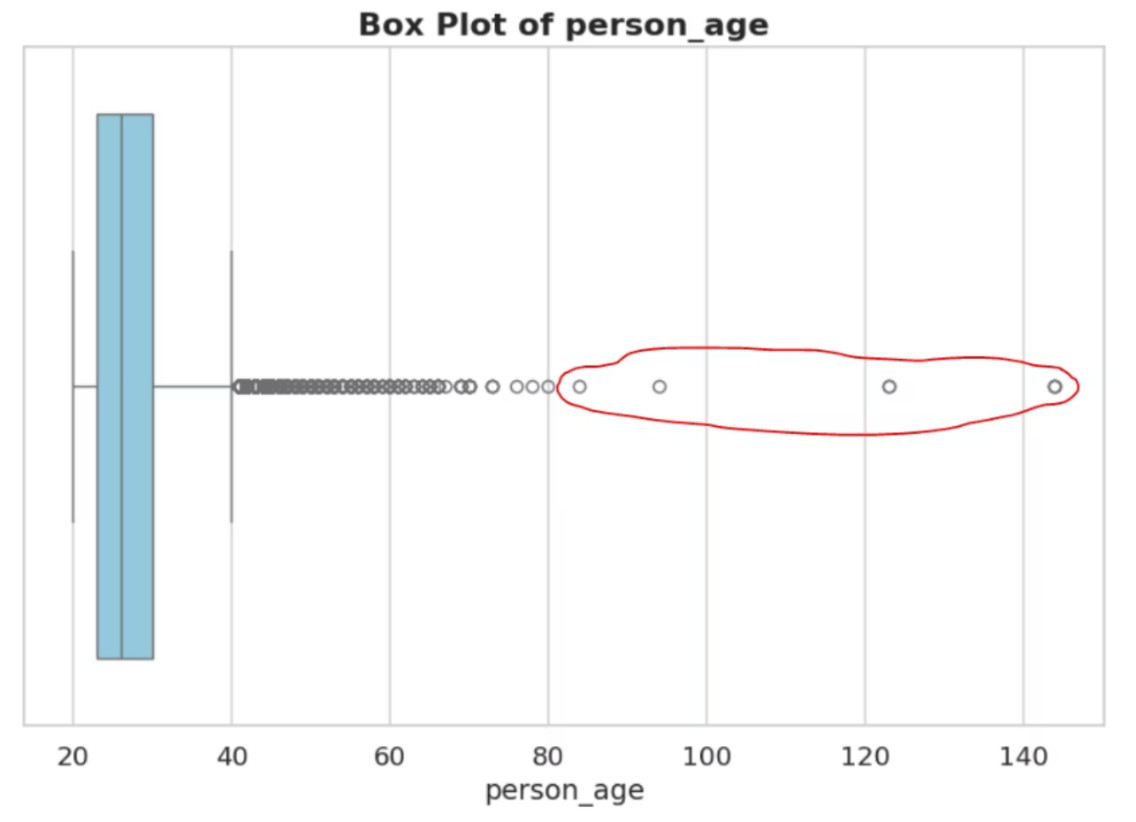
```
# Loop over each numeric column and apply the IQR rule
for col in numeric_cols:
    # Calculate mean and standard deviation
    mean_val = df[col].mean()
    std_val = df[col].std()

    # Define the acceptable range using mean +/- 3 * std deviation
    lower_bound = mean_val - multiplier * std_val
    upper_bound = mean_val + multiplier * std_val

    print ("col=",col)
    print ("lower_bound=",lower_bound)
    print ("upper_bound=",upper_bound)

    # Filter the DataFrame for the current column
    df_clean = df_clean[(df_clean[col] >= lower_bound) & (df_clean[col] <= upper_bound)]
    df_clean.shape[0]
```

Handling Outliers

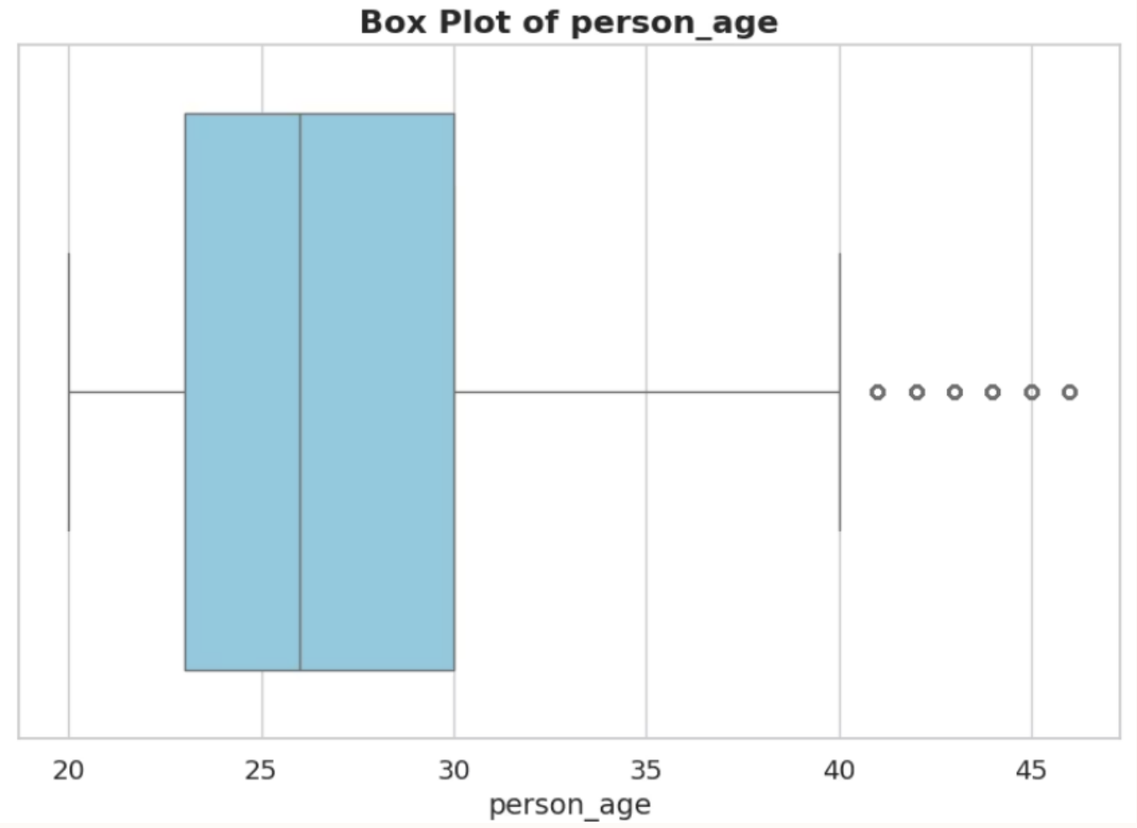
Extracted some variables to demonstrate the effect after removing outliers.

person_age

Before remove outliers



After remove outliers

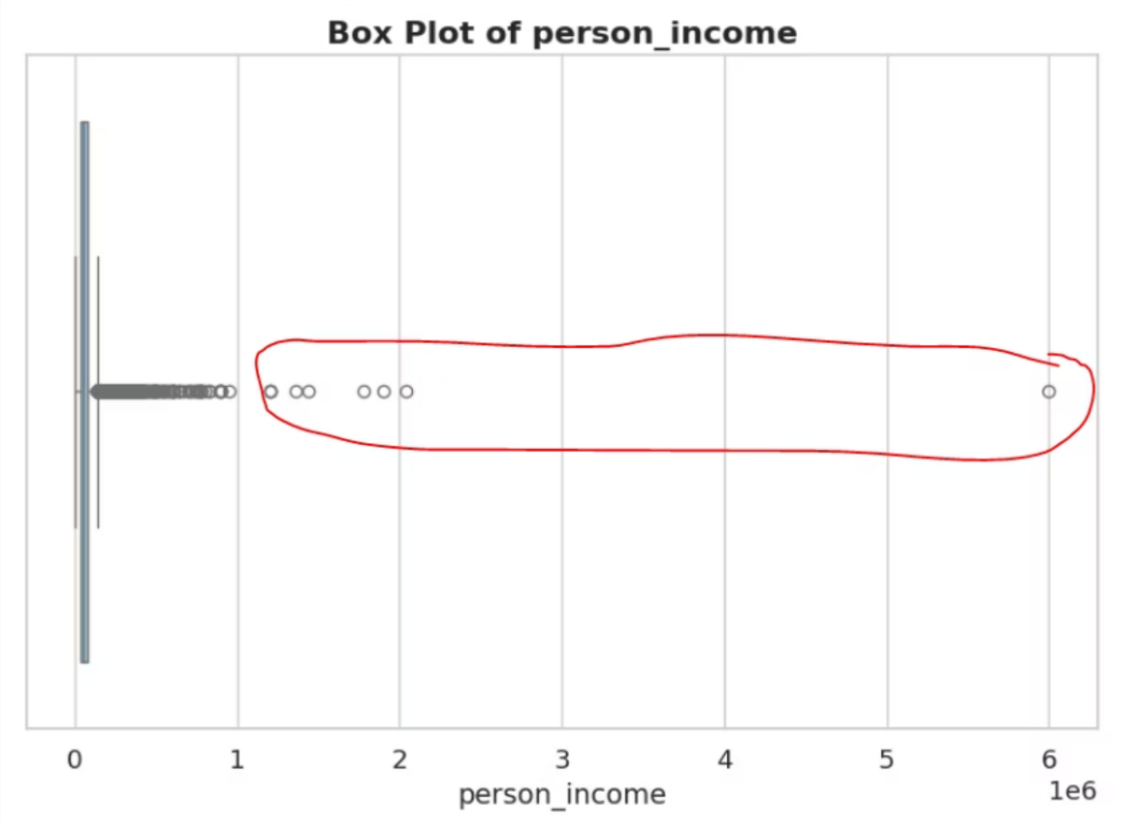


Handling Outliers

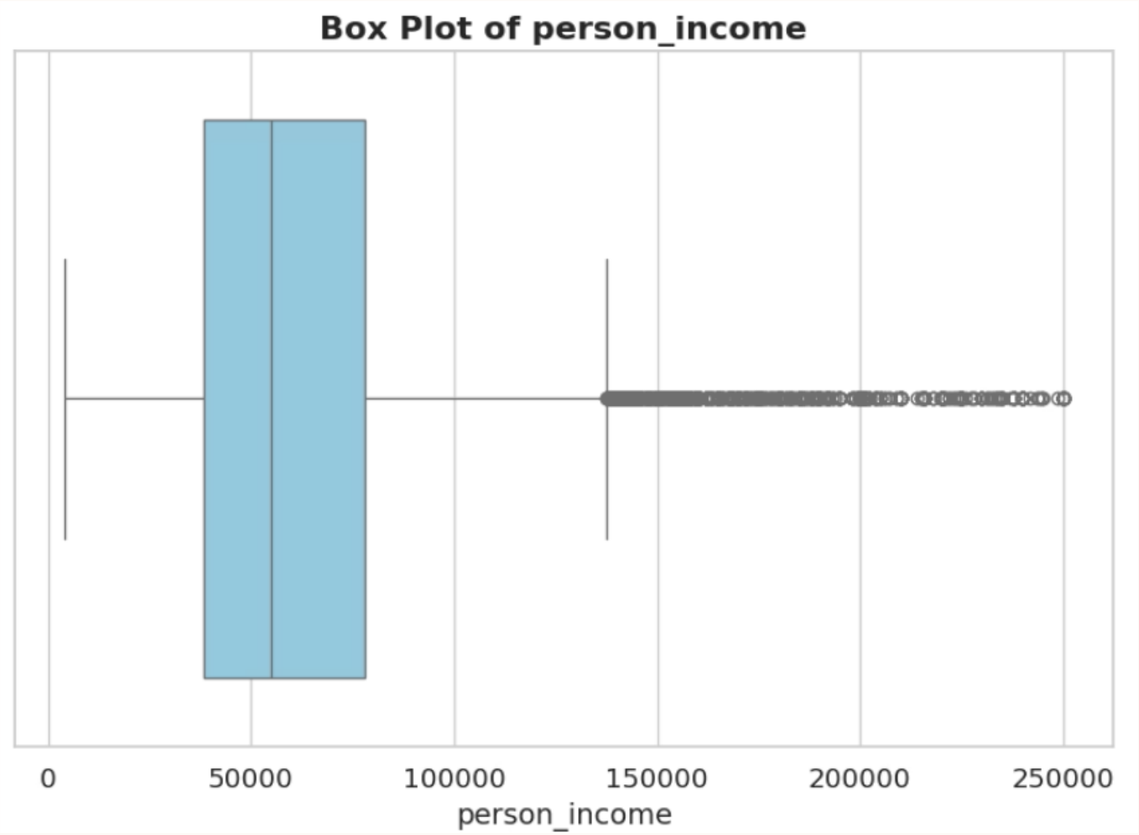
Extracted some variables to demonstrate the effect after removing outliers.

person_income

Before remove outliers



After remove outliers

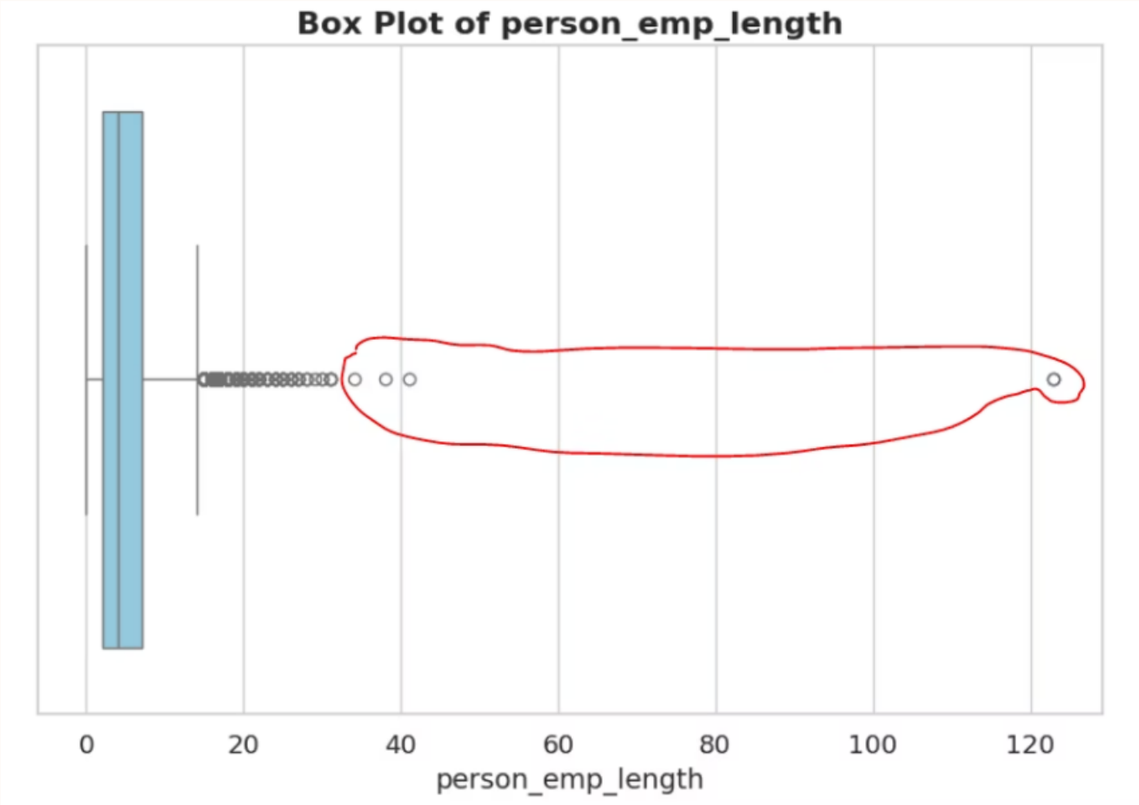


Handling Outliers

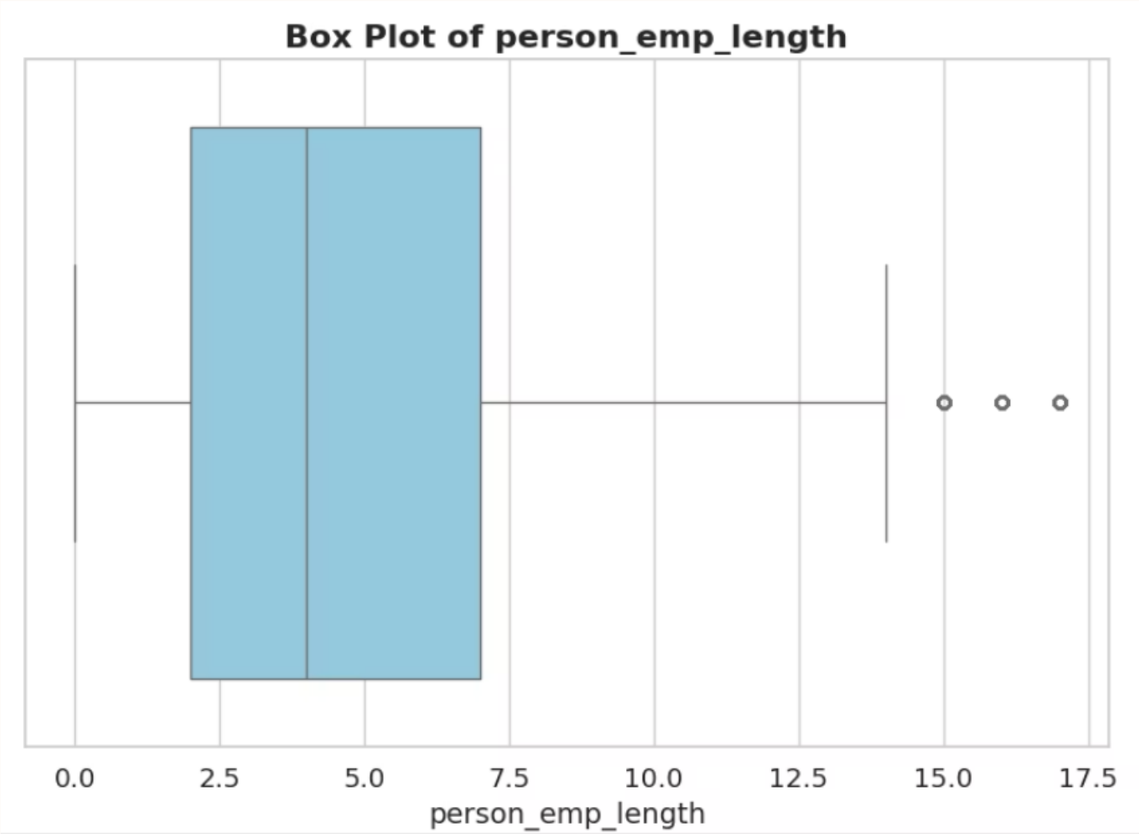
Extracted some variables to demonstrate the effect after removing outliers.

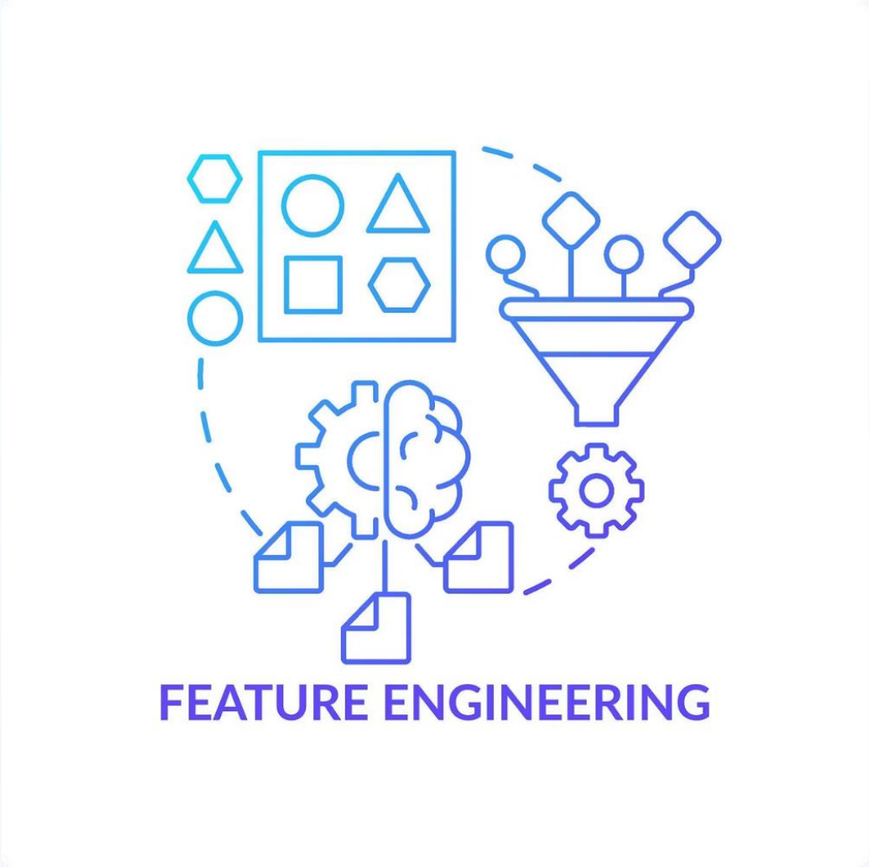
person_emp_length

Before remove outliers



After remove outliers





Feature Engineering

Creating New Features

- In feature engineering, it is often to see if creating derived features enhances predictive power, offers intuitive and interpretable metrics for capturing essential aspects of borrower behavior and financial stability.
- In this project, Credit Age Ratio and Employment Stability were derived

Note: The derived features were created before the data cleansing and therefore it has been gone through the data cleansing steps.

Credit Age Ratio

- This ratio measures the proportion of a person's entire life during which they have had a credit history.

Employment Stability

- This score measures the ratio of the time a person has been employed to their available working years.

One-hot Encoding

person_home_ownership
OWN
OWN
OWN
OWN
MORTGAGE

person_home_ownership_MORTGAGE	person_home_ownership_OTHER	person_home_ownership_OWN	person_home_ownership_RENT
0.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0
0.0	0.0	1.0	0.0
1.0	0.0	0.0	0.0

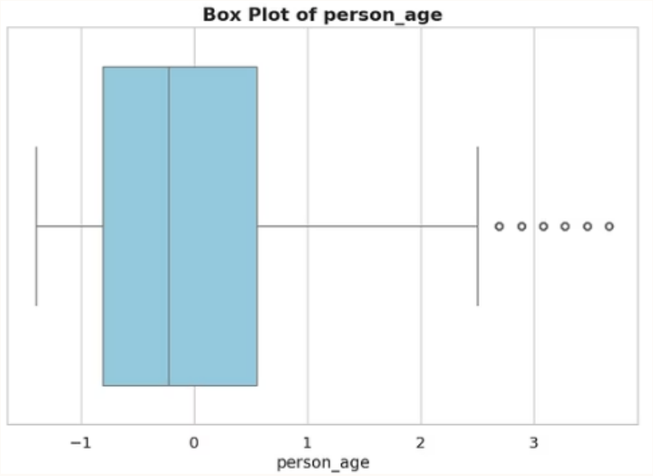
```
# One-Hot Encoding for categorical features (as character could not be processed for decision tree)
encoder = OneHotEncoder(sparse_output=False)  # Use sparse_output instead of sparse
encoded_features = encoder.fit_transform(fullData_no_outliers_reset[char_features])
encoded_features_names = encoder.get_feature_names_out(char_features)
print("Features created from One Hot Encoder:")
print(encoded_features_names)
# Combine the encoded features with the dataset
encoded_df = pd.DataFrame(encoded_features, columns=encoded_features_names)
```

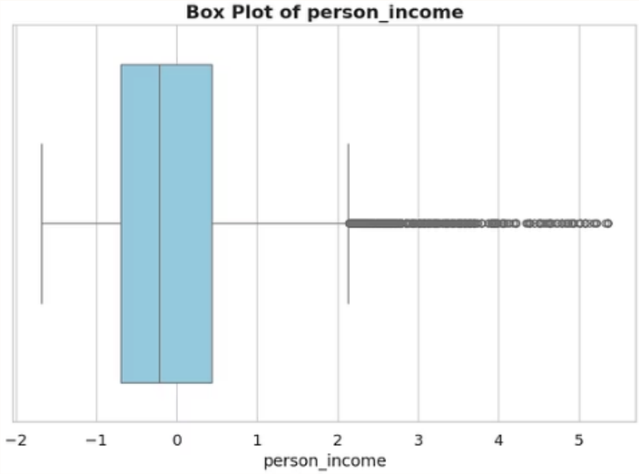
- As many machine learning algorithms process numeric format only
- Character features ('person_home_ownership', 'loan_intent', 'cb_person_default_on_file') were encoded to numbers
- Character fields split into n fields if there is n possible values. Each field is either 0 or 1

Standardization

```
scaler = StandardScaler()
scaled_numeric = scaler.fit_transform(fullData_no_outliers_reset[num_features])
```

After Standardization (examples below)

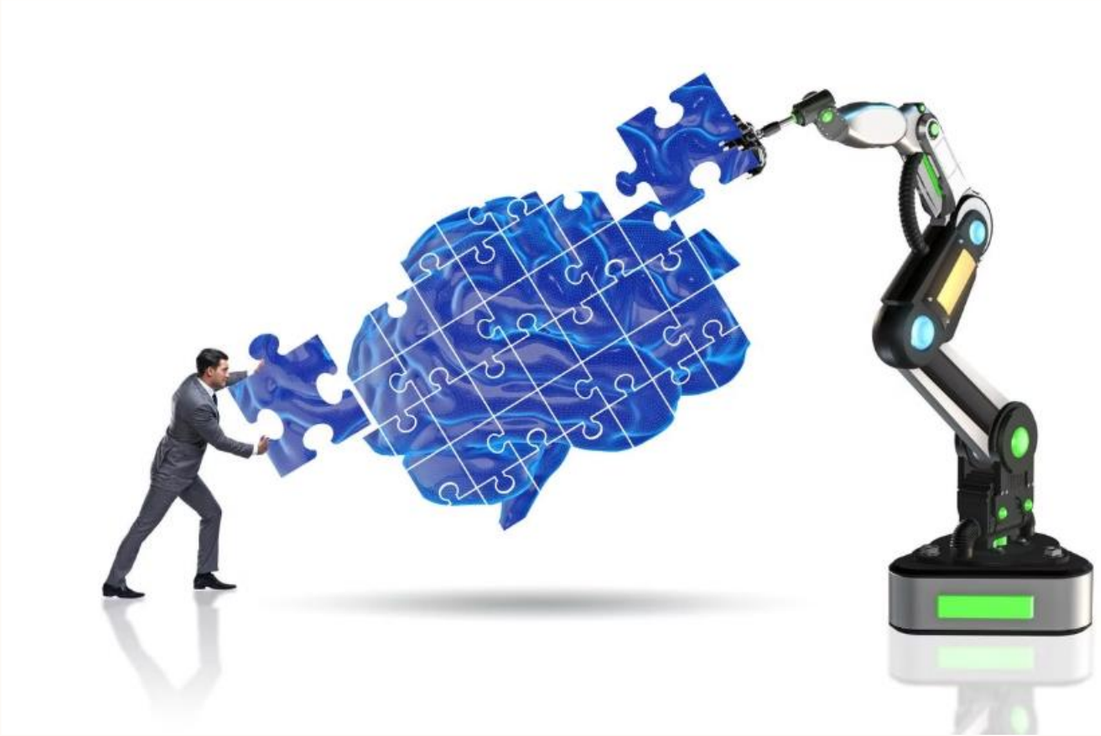




- Some numeric features with large values (e.g. person income) may dominate during the model training
- Do standardization *{Z = (X - mean) / σ}*
- Remove mean and scaling to unit variance (i.e. all feature with mean = 0 and standard deviation = 1)
- All numeric features in same scale

Model Training

- Split sample data to 70% for training and 30% for testing
- Model pre-processing for each algorithm for better performance or model explainability



Model Tuning

Decision Tree

- Do pruning and limit tree depth to 5 and leaf size as 10% of training samples.
- To avoid a very large tree built and lower the model explainability.

Logistic Regression

- Due to model assumptions, remove multicollinearity (VIF > 5) and select features with correlation < 0.1 to increase the model performance.

Random Forest

- Select most important features by feature importance score.
- Correlation was not used for feature selection because correlation would only consider linear relationship and omit non-linear relationship.

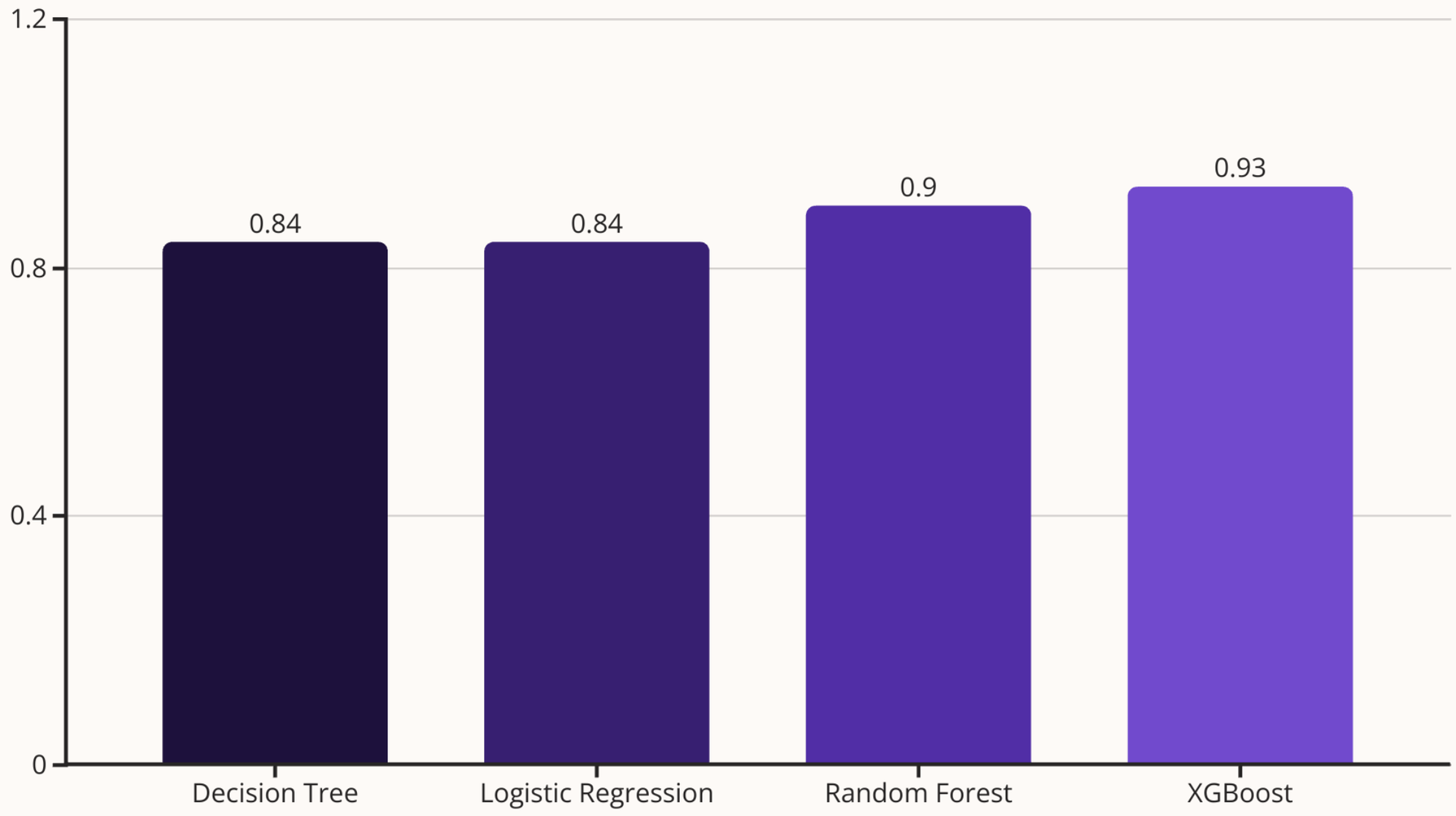
XGBoost

- Hyperparameter tunning: Try several cases on the max_depth, learning_rate, n_estimators, subsample and colsample_bytree and pick the best parameters through cross-validation.

Model Testing

Model Performance Comparison

AUC Results (use 30% of data for testing)



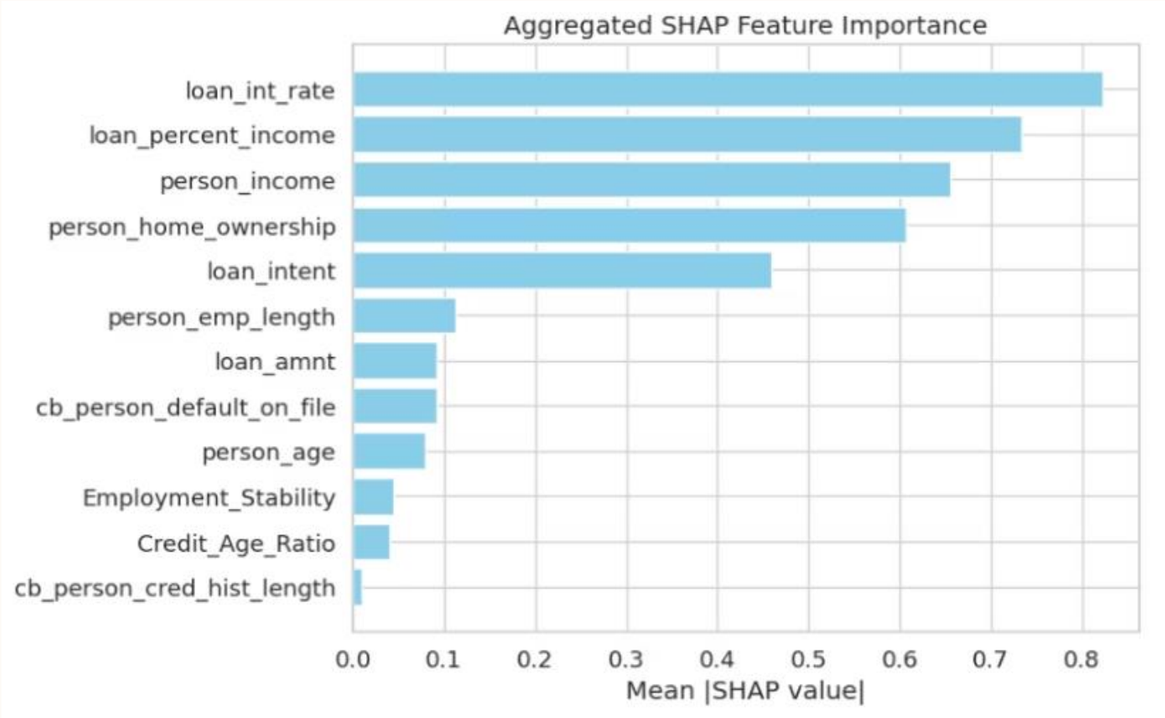
Insight: XGBoost with highest AUC score due to its boosted ensemble approach, offers superior performance.

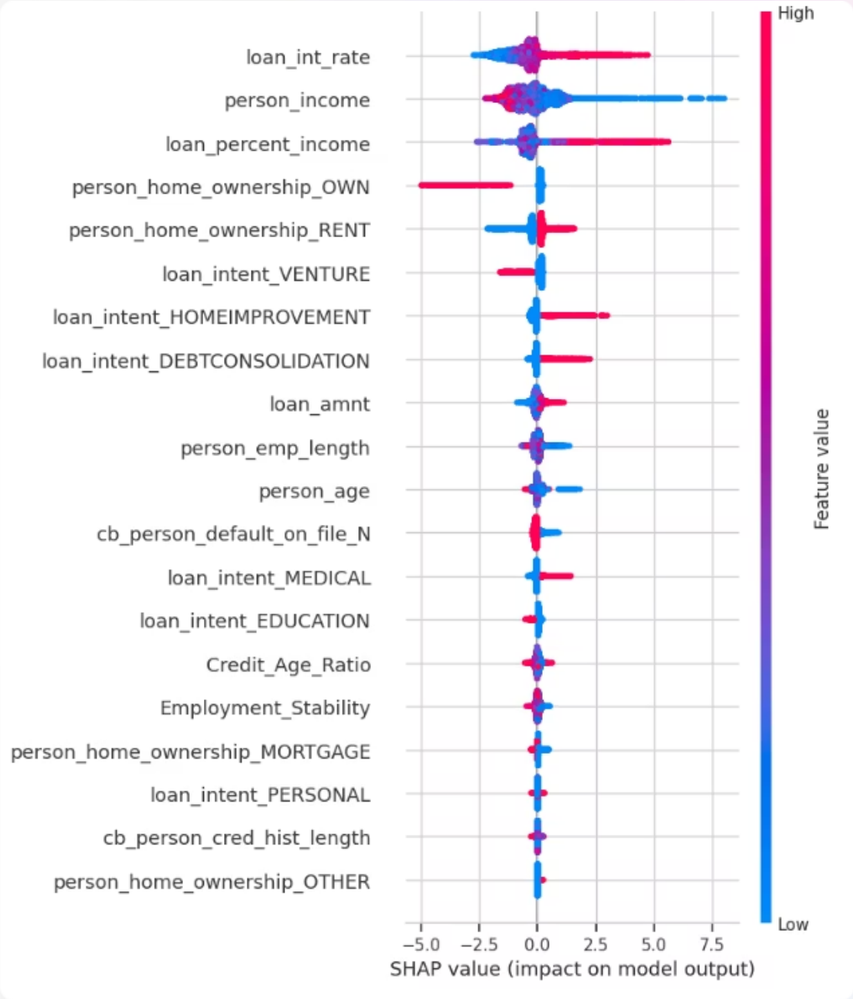
SHAP Feature Importance

- Choose XGBoost model with highest AUC score to move forward to model explainability (SHAP)
- There are character features "person_home_ownership", "loan_intent", "cb_person_default_on_file" which have been applied One Hot Encoding.

- SHAP values in summary plot for One Hot Encoded features are grouped by averaging SHP values to easy investigate the feature importance
- **Observation:**

  - **loan_int_rate, loan_percent_income, person_income, person_home_ownership** and **loan_intent** are features which contribute the most in the model prediction





SHAP Summary Plot

Interpretation on the most important features:

loan_int_rate: Borrower with higher interest rate with higher risk to default

loan_percent_income: Borrower with higher dept-to-income with higher risk to default

person_income: Borrower with higher income provides borrowers more means to cover their expenses and repay loans, hence lowering the risk

person_home_ownership: If a borrower owns their home, there's a strong stabilizing effect on the risk prediction. While renting tends to push the prediction toward higher risk. For other Mortgage and Other, these categories don't seem to contribute meaningful information.

loan_intent: If a borrower is seeking a Venture loan, it is associated with a lower predicted risk. Conversely, if a borrower is seeking Medical loan, it pushes to a higher risk. For other loan intents, they don't seem to contribute meaningful information.

Thank you